

Q.1 Primary Alkyl halide C_4H_9Br (X) reacts with alcoholic KOH to give the compound (Y). Compound (Y) is reacted with HBr to give (Z) which is an isomer of (X). When (X) is reacted with Na/dry ether, it gives a compound (T), C_8H_{18} , which is different from the compound formed when n-butyl bromide is reacted with Na/dry ether. The correct IUPAC name of (T) is:

1. 2,3-Dimethyloctane
2. 2,5-Dimethylhexane
3. 2,3-Diethylbutane
4. 2-Isobutylbutane

Options 1. 1

2. 2
3. 3
4. 4

Q.2 Identify the incorrect statement with reference to biomolecules.

1. In a β -pleated sheet structure, all peptide chains are stretched out to maximum extension and then laid side by side, which are held together by inter-molecular hydrogen bonds.
2. In fibrous proteins, the parallel polypeptide chains are held together by hydrogen and disulphide bonds.
3. Enzymes can catalyze any biochemical reactions.
4. Deficiency of vitamin K increases the blood clotting time.

Options 1. 1

2. 2
3. 3
4. 4

Q.3 Match List-I (compound) with List-II (reagent or chemical test)

List-I	List-II
Compound	Reagent or Chemical test
(A) Phenols	(I) Tollen's test
(B) Amines	(II) Neutral $FeCl_3$ aq. solution
(C) Aldehydes	(III) Iodoform Test
(D) Methyl Ketones	(IV) Hinsberg's Test

Choose the correct answer from the options given below:

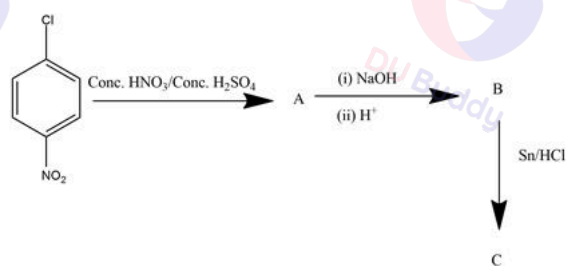
1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
3. (A) - (II), (B) - (I), (C) - (III), (D) - (IV)
4. (A) - (II), (B) - (IV), (C) - (I), (D) - (III)

Options 1. 1

2. 2
3. 3
4. 4

Q.4

Identify A, B, and C in the following reaction:



1. A: 3,4-dinitrochlorobenzene; B: 3,4-dinitrophenol ;C:3,4-diaminophenol
2. A: 2,4-dinitrochlorobenzene; B: 2,4-dinitrophenol ;C:2,4-diaminophenol
3. A: 2,4-dinitrochlorobenzene; B: 2,4-dinitrophenol ; C:2-amino-4-nitrophenol
4. A: 2,4-dinitrochlorobenzene; B: 2,4-dinitrophenol ;C:2-nitro-4-aminophenol

Options 1. 1

2. 2
3. 3
4. 4

Q.5 The correct relation for determining the molar mass of solute (M_2)

using osmotic pressure measurement, is

$$1. M_2 = \frac{w_2^2 RT}{\Pi V}$$

$$2. M_2 = \frac{w_2 RT}{\Pi V}$$

$$3. M_2 = \frac{w_2 RT^2}{\Pi V}$$

$$4. M_2 = \frac{w_2 RT}{\Pi \sqrt{V}}$$

Options 1. 1

2. 2

3. 3

4. 4

Q.6 When aniline is reacted with benzoyl chloride, the correct product is

1. N-Phenylbenzamide

2. Biphenyl

3. p-(Amino)phenylbenzoyl chloride

4. N-Phenylaniline

Options 1. 1

2. 2

3. 3

4. 4

Q.7 When sodium chromate is acidified with sulfuric acid and then further treated with potassium chloride, we obtain...

1. Sodium dichromate

2. Potassium chromate

3. Potassium dichromate

4. Sodium dichromate dihydrate

Options 1. 1

2. 2

3. 3

4. 4

Q.8 With reference to D-glucose, which of the following cannot be explained by its open chain structure?

(A) D-Glucose does not give Schiff's test.

(B) The pentaacetate of D-glucose does not react with hydroxylamine.

(C) Glucose is found to exist in two different crystalline forms.

(D) D-glucose reacts with mild oxidizing agents like $\text{Br}_2\text{-H}_2\text{O}$ and forms gluconic acid.

Choose the correct answer from the options given below:

1. (A), (B), (C) and (D).

2. (A), (B) and (D) only.

3. (B), (C) and (D) only.

4. (A), (B) and (C) only.

Options 1. 1

2. 2

3. 3

4. 4

Q.9 Which of the following methyl halide will have the highest value of dipole moment?

1. $\text{CH}_3\text{-I}$

2. $\text{CH}_3\text{-Br}$

3. $\text{CH}_3\text{-F}$

4. $\text{CH}_3\text{-Cl}$

Options 1. 1

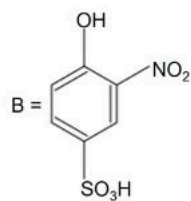
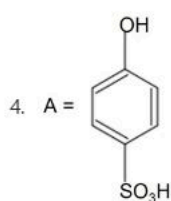
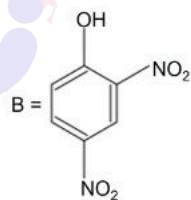
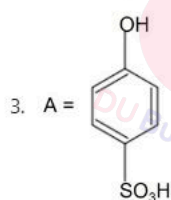
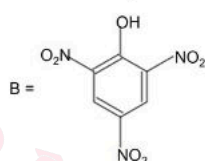
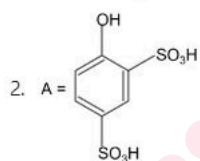
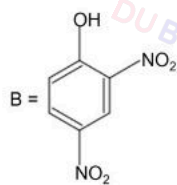
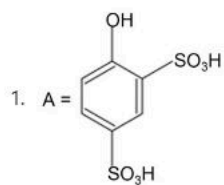
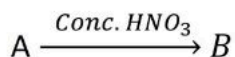
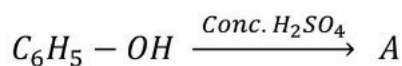
2. 2

3. 3

4. 4

Q.10

Predict the products, A and B in the following reaction sequence:



- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.11 Match List-I with List-II

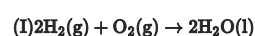
List-I

List-II

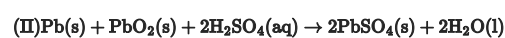
Type of cell/Batteries

Reaction

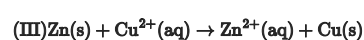
(A) Daniell cell



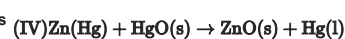
(B) Fuel Cell



(C) Primary Batteries



(D) Secondary Batteries



Choose the correct answer from the options given below:

1. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)
2. (A) - (II), (B) - (I), (C) - (III), (D) - (IV)
3. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
4. (A) - (II), (B) - (IV), (C) - (I), (D) - (III)

- Options
1. 1
 2. 2
 3. 3
 4. 4

Q.12

If the concentration of a reactant is reduced to half, the rate of reaction becomes $\frac{1}{4}$. What will be the order of reaction?

1. 0
2. 1
3. 2
4. 4

Options

1. 1
2. 2
3. 3

Q.13 The nitrogenous base not present in DNA is

1. Adenine
2. Uracil
3. Thymine
4. Guanine

Options 1. 1
2. 2
3. 3
4. 4

Q.14 Arrange the metal ions of the following compounds in increasing order of oxidation number:

- (A) TiCl_3
- (B) CrF_6
- (C) ZnO
- (D) Mn_2O_7

Choose the correct answer from the options given below:

1. (D), (B), (C), (A)
2. (D), (B), (A), (C)
3. (C), (A), (D), (B)
4. (C), (A), (B), (D)

Options 1. 1
2. 2
3. 3
4. 4

Q.15 Which of the following reactions will give n-Propanol as a product?

- (A) Hydroboration oxidation of Propene
- (B) Reduction of propanal with Palladium as a catalyst
- (C) Reaction of Formaldehyde with Ethylmagnesiumbromide followed by hydrolysis
- (D) Acid catalyzed hydration of propene

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (A), (C) and (D) only

Options 1. 1
2. 2
3. 3
4. 4

Q.16 The correct increasing order of reactivity of alkyl halides with CH_3NH_2 is

- (A) $\text{C}_2\text{H}_5\text{-Br}$
- (B) $\text{C}_2\text{H}_5\text{-I}$
- (C) $\text{C}_2\text{H}_5\text{-F}$
- (D) $\text{C}_2\text{H}_5\text{-Cl}$

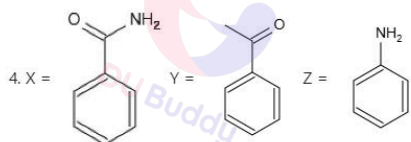
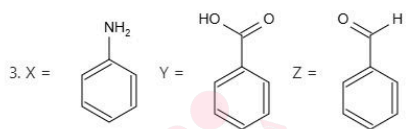
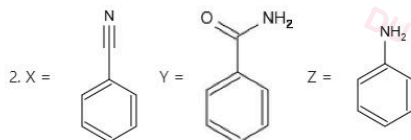
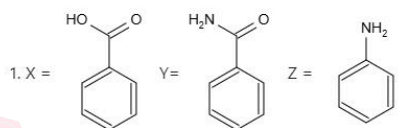
Choose the correct answer from the options given below:

1. (A), (B), (D), (C)
2. (C), (D), (A), (B)
3. (C), (D), (B), (A)
4. (B), (A), (D), (C)

Options 1. 1
2. 2
3. 3
4. 4

Q.17

An aromatic compound 'X' on treatment with aqueous ammonia and heating forms the compound 'Y' which on heating with Br_2 and KOH forms a compound 'Z' of the molecular formula $\text{C}_6\text{H}_7\text{N}$. The structures of X, Y and Z can be



- Options
- 1
 - 2
 - 3
 - 4

Q.18 Calculate the magnetic moment (spin-only) of a trivalent ion in aqueous solution if its atomic number is 22.

1. 1.44 BM
2. 1.73 BM
3. 2.84 BM
4. 0

- Options
- 1
 - 2
 - 3
 - 4

Q.19 From binary solution of two liquids, A and B

Match List-I with List-II

List-I: Inter-molecular interaction

List-II: Behavior

(A). A-A & B-B are nearly equal to A-B

(I). Negative deviation from Raoult's Law

(B). A-A & B-B are stronger compared to A-B

(II). Azeotropes

(C). A-A & B-B are weaker compared to A-B

(III). Positive deviation from Raoult's Law

(D). Very strong A-B interaction

(IV). No deviation from Raoult's Law

Choose the correct answer from the options given below:

1. (A) - (IV), (B) - (III), (C) - (I), (D) - (II)
2. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (II), (C) - (IV), (D) - (I)

- Options
- 1
 - 2
 - 3
 - 4

Q.20 The conductivity of an electrolytic solution does not depend on

1. Osmotic pressure
2. Concentration of the electrolyte
3. The nature of electrolyte
4. Temperature

- Options
- 1
 - 2
 - 3

- Q.21 The correct sequence of steps during the corrosion of iron is
- (A) Ferrous ions are oxidized by atmospheric oxygen
 - (B) Hydrated ferric oxide ($\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$) formed with production of H^+ ions
 - (C) At a particular spot of iron object, oxidation takes place and that spot behaves as anode
 - (D) Electron released at anode moves through metal and reduces oxygen in the presence of H^+

Choose the correct answer from the options given below:

1. (A), (C), (B), (D)
2. (C), (D), (B), (A)
3. (C), (B), (D), (A)
4. (C), (D), (A), (B)

Options 1. 1

2. 2
3. 3
4. 4

- Q.22 An increase in the density is seen when moving from Ti to Cu in transition metal series. This can be attributed to

1. lanthanoid contraction
2. filling of f orbitals prior to d orbitals
3. Ineffective shielding of d electrons
4. Increase in atomic number and atomic size

Options 1. 1

2. 2
3. 3
4. 4

- Q.23 Which biomolecule is an example of fibrous protein?

1. Epinephrine
2. Insulin
3. Albumin
4. Myosin

Options 1. 1

2. 2
3. 3
4. 4

- Q.24 Question 4 (IUPAC Name of Freon 12):

The correct IUPAC name of Freon 12 is

1. 1,1-Difluoro-1,1-dichloromethane
2. 1,1-Dichloro-1,1-difluoromethane
3. 1,1-Difluoro-2,2-dichloroethane
4. 1,1-Dichloro-2,2-difluoroethane

Options 1. 1

2. 2
3. 3
4. 4

- Q.25 Which of the following is a characteristic of $\text{S}_\text{N}2$ reaction?

- (A). Concentration of both reactants determines the reaction rate.
- (B). Proceeds with inversion of configuration.
- (C). The transition state involves central carbon bonded to five species.
- (D). Occurs in polar protic solvent

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

- Q.26 Arrange the following in the decreasing order of basic strength in gas phase:

- (A) $\text{C}_2\text{H}_5\text{NH}_2$
- (B) $(\text{C}_2\text{H}_5)_2\text{NH}$
- (C) $(\text{C}_2\text{H}_5)_3\text{N}$
- (D) NH_3

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (B), (A), (C), (D)
3. (B), (A), (D), (C)
4. (C), (B), (A), (D)

Options 1. 1

2. 2

3. 3
4. 4

Q.27 Which of the following ethers cannot be prepared by bimolecular dehydration of alcohol?

- (A) 2-Methoxy-2-methylpropane
(B) 2-Methoxy propane
(C) 1-Propoxypropane
(D) 2-Ethoxy butane

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1
2. 2
3. 3
4. 4

Q.28 Match List-I with List-II

List-I: Reaction property

(A) Half-life of decomposition of gaseous ammonia on hot platinum surface

(B) Instantaneous rate

(C) Rate constant of inversion of cane sugar

(D) Half-life of acid hydrolysis of methyl acetate

List-II: Formula

(I) $\frac{1}{t} \ln \left(\frac{[A]_0}{[A]_t} \right)$

(II) $\frac{[R]_0}{2k}$

(III) $-\frac{dx}{dt}$

(IV) $\frac{\ln(2)}{k}$

Choose the correct answer from the options given below:

1. (A) - (II), (B) - (I), (C) - (III), (D) - (IV)
2. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
3. (A) - (II), (B) - (III), (C) - (I), (D) - (IV)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1
2. 2
3. 3
4. 4

Q.29 Arrange the following in the increasing order of their boiling points

- (A) $n\text{-C}_4\text{H}_9\text{NH}_2$
(B) $(\text{C}_2\text{H}_5)_2\text{NH}$
(C) $n\text{-C}_4\text{H}_9\text{OH}$
(D) $\text{C}_2\text{H}_5\text{N}(\text{CH}_3)_2$

Choose the correct answer from the options given below:

1. (A), (C), (B), (D)
2. (D), (B), (C), (A)
3. (B), (A), (D), (C)
4. (D), (B), (A), (C)

Options 1. 1
2. 2
3. 3
4. 4

Q.30 Λ_m° for NaCl, HCl, and NaAc are 126.4, 425.9 and 91.0 $\text{S cm}^2\text{mol}^{-1}$, respectively. Λ_m° for HAc is

1. 643.3 $\text{S cm}^2\text{mol}^{-1}$
2. 390.5 $\text{S cm}^2\text{mol}^{-1}$
3. 208.5 $\text{S cm}^2\text{mol}^{-1}$
4. 461.3 $\text{S cm}^2\text{mol}^{-1}$

Options 1. 1
2. 2
3. 3
4. 4

Q.31 From the following statement regarding the order of reaction, the statement that is not true is

1. Order of reaction is an experimental quantity
2. The order of a reaction can be zero or even a fraction
3. Order of reaction is applicable to elementary reaction
4. For complex reaction, order is given by the fastest step

Options 1. 1
2. 2
3. 3
4. 4

Q.32 Azeotropes are binary mixtures having

1. Same composition in liquid and vapor phase, constant boiling temperature and they cannot be separated by fractional distillation.

- The same composition in the liquid and vapor phase, constant boiling temperature and they can be separated by distillation
- The same composition in the liquid and vapor phase, constant boiling temperature, they cannot be separated by fractional distillation and have equal volume of the two liquids.
- Constant boiling temperature, they can be separated by fractional distillation and have equal volumes of two liquids.

Options 1. 1

- 2
- 3
- 4

Q.33 Match List-I (biomolecule) with List-II (characteristics)

List-I: Biomolecule List-II: Characteristics

- | | |
|-------------------|--------------------------|
| (A) Lactose | (I) Menstrual cycle |
| (B) Glycine | (II) Bleeding gums |
| (C) Ascorbic acid | (III) Optically inactive |
| (D) Estradiol | (IV) Reducing sugar |

Choose the correct answer from the options given below:

- (A) - (IV), (B) - (III), (C) - (I), (D) - (II)
- (A) - (III), (B) - (IV), (C) - (II), (D) - (I)
- (A) - (III), (B) - (IV), (C) - (I), (D) - (II)
- (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

Options 1. 1

- 2
- 3
- 4

Q.34 Phenols are acidic in nature. Which of the following statements is incorrect about them?

- Phenols are soluble in aqueous alkali solution
- Phenol liberates hydrogen gas on reaction with sodium metal.
- Phenol liberates carbon dioxide on reaction with sodium bicarbonate.
- Phenol is less stabilized by resonance than phenoxide ion.

Options 1. 1

- 2
- 3
- 4

Q.35 A solution of CuSO_4 is electrolyzed for 1 minute with a current of 1 ampere. Given the atomic mass of Cu as 63 g mol^{-1} , the mass of copper deposited at the cathode is

- $(945/F) \text{ g}$
- $(1890/F) \text{ g}$
- $(3780/F) \text{ g}$
- $(4725/F) \text{ g}$

Options 1. 1

- 2
- 3
- 4

Q.36 Actinoid contraction is greater from element to element than Lanthanoid contraction because of

- poor shielding by $4f$ electrons
- prior filling of $4d$ electrons
- prior filling of $3d$ electrons
- poor shielding by $5f$ electrons

Options 1. 1

- 2
- 3
- 4

Q.37 The flow of solvent from pure solvent side to solution side across a semipermeable membrane can be stopped if some extra pressure is applied on the solution. This pressure that just stops the flow of solvent is called

- Osmotic pressure of solution
- Osmotic pressure of solvent
- Reverse Osmotic pressure of solution
- Reverse Osmotic pressure of solvent

Options 1. 1

- 2
- 3
- 4

- Q.38 Choose the correct statements from the following regarding the reaction kinetics
- (A) A catalyst does not affect the Gibbs energy of a reaction.
 - (B) A catalyst alters the equilibrium constant.
 - (C) For a zero order reaction, half life is dependent on the initial concentration of the reactant.
 - (D) Orientation of reactant molecules affects the reaction rate.
- Choose the correct answer from the options given below:
1. (A), (B) and (D) only
 2. (A), (B) and (C) only
 3. (A), (C) and (D) only
 4. (B), (C) and (D) only

Options 1. 1
2. 2
3. 3
4. 4

- Q.39 The van't Hoff Factor, i for complete dissociation of (i) NaCl, (ii) MgSO_4 and (iii) K_2SO_4 are
1. (i) 1, (ii) 2, (iii) 3
 2. (i) 2, (ii) 2, (iii) 3
 3. (i) 3, (ii) 2, (iii) 1
 4. (i) 2, (ii) 3, (iii) 3

Options 1. 1
2. 2
3. 3
4. 4

- Q.40 A catalyst increases the rate of reaction by:
1. Decreasing the Gibbs energy with unchanged equilibrium state
 2. Increasing the Gibbs energy and changing the equilibrium state
 3. Decreasing the activation energy with unchanged equilibrium state
 4. Decreasing the activation energy and changing the equilibrium state

Options 1. 1
2. 2
3. 3
4. 4

- Q.41 Read the passage carefully and answer the Questions

In the formation of coordination complexes, if the inner d orbital $(n-1)d$ is used in hybridisation, the complex, is called an inner orbital or low spin or spin paired complex. And if it uses outer d orbital (nd) in hybridisation (like sp^3d^2), it is called outer orbital or high spin or spin free complex. The degeneracy of the d orbitals has been removed due to ligand electron-metal electron repulsions in the complex. This splitting of the degenerate levels due to the presence of ligands in a definite geometry is termed as crystal field splitting. For coordination complexes, the magnetic moment is determined by the number of unpaired electrons and is calculated by using the 'spin-only' formula, $\mu = \sqrt{n(n+2)}$ where n is the number of unpaired electrons and μ is the magnetic moment in units of Bohr magneton (BM). The coordination compounds are of great importance. These compounds are widely present in the mineral, plant and animal worlds and are known to play many important functions in the area of analytical chemistry, metallurgy, biological systems, industry and medicine.

Which of the following complex has the highest magnetic moment?

1. $[\text{Mn}(\text{CN})_6]^{3-}$
2. $[\text{FeF}_6]^{3-}$
3. $[\text{CoF}_6]^{3-}$
4. $[\text{MnCl}_6]^{3-}$

Options 1. 1
2. 2
3. 3
4. 4

- Q.42 Read the passage carefully and answer the Questions

In the formation of coordination complexes, if the inner d orbital $(n-1)d$ is used in hybridisation, the complex, is called an inner orbital or low spin or spin paired complex. And if it uses outer d orbital (nd) in hybridisation (like sp^3d^2), it is called outer orbital or high spin or spin free complex. The degeneracy of the d orbitals has been removed due to ligand electron-metal electron repulsions in the complex. This splitting of the degenerate levels due to the presence of ligands in a definite geometry is termed as crystal field splitting. For coordination complexes, the magnetic moment is determined by the number of unpaired electrons and is calculated by using the 'spin-only' formula, $\mu = \sqrt{n(n+2)}$ where n is the number of unpaired electrons and μ is the magnetic moment in units of Bohr magneton (BM). The coordination compounds are of great importance. These compounds are widely present in the mineral, plant and animal worlds and are known to play many important functions in the area of analytical chemistry, metallurgy, biological systems, industry and medicine.

$[\text{Co}(\text{NH}_3)_6]^{3+}$ and $[\text{Ni}(\text{NH}_3)_6]^{2+}$ respectively are

1. Both inner orbital complexes
2. Both outer orbital complexes
3. Outer orbital complex, Inner orbital complex
4. Inner orbital complex, outer orbital complex

Options 1. 1
2. 2
3. 3
4. 4

Q.43 Read the passage carefully and answer the Questions

In the formation of coordination complexes, if the inner d orbital $(n-1)d$ is used in hybridisation, the complex, is called an inner orbital or low spin or spin paired complex. And if it uses outer d orbital (nd) in hybridisation (like sp^3d^2), it is called outer orbital or high spin or spin free complex. The degeneracy of the d orbitals has been removed due to ligand electron-metal electron repulsions in the complex. This splitting of the degenerate levels due to the presence of ligands in a definite geometry is termed as crystal field splitting. For coordination complexes, the magnetic moment is determined by the number of unpaired electrons and is calculated by using the 'spin-only' formula, $\mu = \sqrt{n(n+2)}$ where n is the number of unpaired electrons and μ is the magnetic moment in units of Bohr magneton (BM). The coordination compounds are of great importance. These compounds are widely present in the mineral, plant and animal worlds and are known to play many important functions in the area of analytical chemistry, metallurgy, biological systems, industry and medicine.

The CFSE for tetrahedral $[\text{CoCl}_4]^{2-}$ is 8000 cm^{-1} , the CFSE for octahedral $[\text{CoCl}_6]^{4-}$ will be.....

1. 16000 cm^{-1}
2. 3555.2 cm^{-1}
3. 8000 cm^{-1}
4. 18000 cm^{-1}

Options 1. 1

2. 2
3. 3
4. 4

Q.44 Read the passage carefully and answer the Questions

In the formation of coordination complexes, if the inner d orbital $(n-1)d$ is used in hybridisation, the complex, is called an inner orbital or low spin or spin paired complex. And if it uses outer d orbital (nd) in hybridisation (like sp^3d^2), it is called outer orbital or high spin or spin free complex. The degeneracy of the d orbitals has been removed due to ligand electron-metal electron repulsions in the complex. This splitting of the degenerate levels due to the presence of ligands in a definite geometry is termed as crystal field splitting. For coordination complexes, the magnetic moment is determined by the number of unpaired electrons and is calculated by using the 'spin-only' formula, $\mu = \sqrt{n(n+2)}$ where n is the number of unpaired electrons and μ is the magnetic moment in units of Bohr magneton (BM). The coordination compounds are of great importance. These compounds are widely present in the mineral, plant and animal worlds and are known to play many important functions in the area of analytical chemistry, metallurgy, biological systems, industry and medicine.

Fe(II) octahedral complexes can show different 'spin only' magnetic moments for arrangements in the presence of weak and strong ligands. The 'spin only' magnetic moments of these complexes with weak and strong ligands, respectively, are

1. 1.0 BM and 4.90 BM
2. 4.90 BM and 0 BM
3. 5.92 BM and 1.73 BM
4. 1.73 BM and 5.92 BM

Options 1. 1

2. 2
3. 3
4. 4

Q.45 Read the passage carefully and answer the Questions

In the formation of coordination complexes, if the inner d orbital $(n-1)d$ is used in hybridisation, the complex, is called an inner orbital or low spin or spin paired complex. And if it uses outer d orbital (nd) in hybridisation (like sp^3d^2), it is called outer orbital or high spin or spin free complex. The degeneracy of the d orbitals has been removed due to ligand electron-metal electron repulsions in the complex. This splitting of the degenerate levels due to the presence of ligands in a definite geometry is termed as crystal field splitting. For coordination complexes, the magnetic moment is determined by the number of unpaired electrons and is calculated by using the 'spin-only' formula, $\mu = \sqrt{n(n+2)}$ where n is the number of unpaired electrons and μ is the magnetic moment in units of Bohr magneton (BM). The coordination compounds are of great importance. These compounds are widely present in the mineral, plant and animal worlds and are known to play many important functions in the area of analytical chemistry, metallurgy, biological systems, industry and medicine.

Which of the following metals belongs to Wilkinson catalyst?

1. Rh
2. Pt
3. Pd
4. Ti + Al

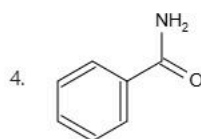
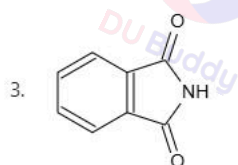
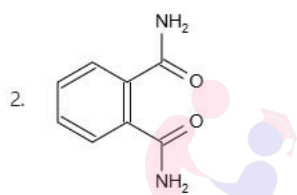
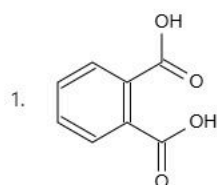
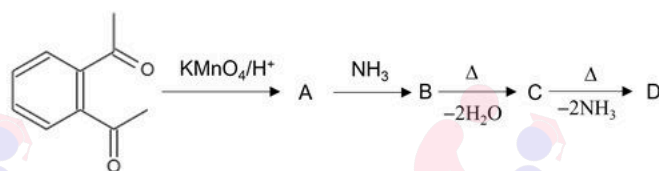
Options 1. 1

2. 2
3. 3
4. 4

Q.46 Read the passage carefully and answer the Questions

Aldehydes and ketones are important organic compounds prepared mainly by oxidation of alcohols. Primary alcohols give aldehyde and secondary alcohols yield ketones. Aldehydes can also be formed by partial reduction of acid chlorides, reduction of esters or by Gattermann-Koch reaction. Ketones are prepared by Friedel-Craft's acylation, hydration of alkyne and reaction of Grignard's reagent with nitriles. Chemically, aldehydes and ketones are reactive in nature due to the carbonyl group and undergo nucleophilic addition reactions. Aldehydes are more reactive than ketones and can be oxidized to carboxylic acids, while both can be reduced to alcohols or hydrocarbons and exhibit condensation reaction.

Predict the end product (D) of the following reaction sequence;



Options 1. 1

2. 2

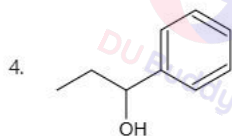
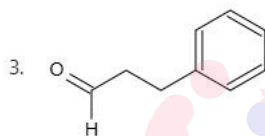
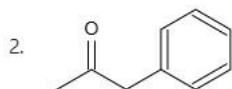
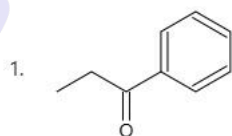
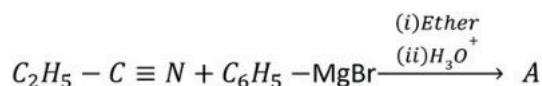
3. 3

4. 4

Q.47 Read the passage carefully and answer the Questions

Aldehydes and ketones are important organic compounds prepared mainly by oxidation of alcohols. Primary alcohols give aldehyde and secondary alcohols yield ketones. Aldehydes can also be formed by partial reduction of acid chlorides, reduction of esters or by Gattermann-Koch reaction. Ketones are prepared by Friedel-Craft's acylation, hydration of alkyne and reaction of Grignard's reagent with nitriles. Chemically, aldehydes and ketones are reactive in nature due to the carbonyl group and undergo nucleophilic addition reactions. Aldehydes are more reactive than ketones and can be oxidized to carboxylic acids, while both can be reduced to alcohols or hydrocarbons and exhibit condensation reaction.

The correct structure of the product in the following reaction is;



Options 1. 1

2. 2

3. 3

4. 4

Q.48 Read the passage carefully and answer the Questions

Aldehydes and ketones are important organic compounds prepared mainly by oxidation of alcohols. Primary alcohols give aldehyde and secondary alcohols yield ketones. Aldehydes can also be formed by partial reduction of acid chlorides, reduction of esters or by Gattermann-Koch reaction. Ketones are prepared by Friedel-Craft's acylation, hydration of alkyne and reaction of Grignard's reagent with nitriles. Chemically, aldehydes and ketones are reactive in nature due to the carbonyl group and undergo nucleophilic addition reactions. Aldehydes are more reactive than ketones and can be oxidized to carboxylic acids, while both can be reduced to alcohols or hydrocarbons and exhibit condensation reaction.

When two moles of benzaldehyde are heated with concentrated NaOH solution, the product/s obtained is/are:

1. Benzyl alcohol and sodium benzoate.
2. Benzyl alcohol only.
3. Sodium benzoate only.
4. Benzophenone only.

Options 1. 1

2. 2

3. 3

4. 4

Q.49 Read the passage carefully and answer the Questions

Aldehydes and ketones are important organic compounds prepared mainly by oxidation of alcohols. Primary alcohols give aldehyde and secondary alcohols yield ketones. Aldehydes can also be formed by partial reduction of acid chlorides, reduction of esters or by Gattermann-Koch reaction. Ketones are prepared by Friedel-Craft's acylation, hydration of alkyne and reaction of Grignard's reagent with nitriles. Chemically, aldehydes and ketones are reactive in nature due to the carbonyl group and undergo nucleophilic addition reactions. Aldehydes are more reactive than ketones and can be oxidized to carboxylic acids, while both can be reduced to alcohols or hydrocarbons and exhibit condensation reaction.

The correct IUPAC name of Vanillin is

1. 4-Hydroxy-3-methoxybenzaldehyde
2. 3-Methoxy-4-hydroxybenzaldehyde
3. 3-Hydroxy-4-methoxybenzaldehyde
4. 4-Methoxy-3-hydroxybenzaldehyde

Options 1. 1

2. 2

3. 3

4. 4

Q.50 Read the passage carefully and answer the Questions

Aldehydes and ketones are important organic compounds prepared mainly by oxidation of alcohols. Primary alcohols give aldehyde and secondary alcohols yield ketones. Aldehydes can also be formed by partial reduction of acid chlorides, reduction

of esters or by Gattermann-Koch reaction. Ketones are prepared by Friedel-Craft's acylation, hydration of alkyne and reaction of Grignard's reagent with nitriles. Chemically, aldehydes and ketones are reactive in nature due to the carbonyl group and undergo nucleophilic addition reactions. Aldehydes are more reactive than ketones and can be oxidized to carboxylic acids, while both can be reduced to alcohols or hydrocarbons and exhibit condensation reaction.

Hydration of propyne in the presence of H_2SO_4 and HgSO_4 gives

1. Propan-1-al
2. Propan-2-one
3. Propan-1-ol
4. Propan-2-ol

Options 1. 1

2. 2

3. 3

4. 4